

MATH2007A
Elementary Calculus 2 Summer 2026

Instructor: Mathieu Lemire
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Lectures: Our lectures are on Mondays and Wednesdays from 18:35 to 19:55 at room 210 Tory Building. Our first class is on Wednesday May 6th.

Tutorials: Tutorials are scheduled to be on Mondays from 17:35 to 18:25. The first tutorial is on Monday May 11th. The following table give more details:

Section	Room	TA's name	TA's connect email
A1	210 TB	Banafsheh Sadeghi Kenarsari	BanafshehSadeghiKena@cmail.carleton.ca
A2	202 TB	Vandana Rathee	VANDANARATHEE@cmail.carleton.ca

Office hours: After our classes on Mondays and Wednesdays. Also, if you want, you can write to me and we can schedule a virtual meeting on Zoom.

Textbook: Single Variable Calculus: Early Transcendentals, 9th edition (you can also used the 8th edition), by James Stewart, Brooks/Cole. The book is available for online purchase.

Prerequisites: MATH1004 or a grade of C- or higher in MATH1007; or permission of the school.

Evaluation: Your final grade will be calculated as:

$$\text{Term Mark } 60 \% (\text{best } 3 \text{ out of } 4 \text{ tests, } 20 \% \text{ each}) + \text{Final Examination } 40\%$$

Term Mark: There will be four tests during the tutorials on **June 1st, June 15th, July 13th and July 27th. No make up, early or delayed tests will be given. The tests will each be 50 minutes long. You can miss a test for free since we only count the best 3 out of 4 tests. Otherwise it is ONLY if you miss a second test that I will need to see a medical note.**

Final exam: The final exam is a cumulative three hours closed book exam scheduled by the university. The exam period runs from August 17th to August 23rd (including Saturdays). It is student's responsibility to be available at the time of the examination. Students who missed the final examination may be eligible for a deferred exam provided that they present a doctor note or another supporting document to the Registrars Office. It is the Registrars Office and not the instructor which take decision of granting a deferred examination. After the exam is written, students may see their final examination papers. This examination review is for educational purpose only and NOT for negotiation of the grade.

Calculators: Only non-programmable and non-graphical calculators are allowed for tests and the final exam.

Practice problems lists Practice problems lists will regularly be posted on Brightspace. These problems are not to be handed in and will not be graded. However, in order to succeed in the course, it is absolutely essential to practice on a regular basis. The practice questions will be taken from the textbook for the course.

Withdrawal: The last day for academic withdrawal is **August 1st.**

Students with Disabilities: Students with disabilities who require academic accommodations in this course are encouraged to contact the Paul Menton Centre for Students with Disabilities to complete the

necessary Letters of Accommodation. After registering with the PMC, make an appointment to meet with me and discuss your needs in order to make the necessary arrangements as early in the term as possible. Please consult the PMC website for the deadline to request accommodations for the formally-scheduled exam (if applicable).

Notes:

1. Problem lists, comments, solutions and other informations will regularly be posted on cuLearn. It is your responsibility to look on Brightspace to obtain these informations.

2. **I will not necessary follow the same order of topics as in the textbook. The best way to know where exactly we are in class is to come to class or to follow the order of topics found in the practice problems lists.**

3. Coming to class is very important and I strongly encourage you to do so.

4. **Pregnancy accommodation:** write to me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details visit the Equity Services webpage.

5. **Religious obligation:** write to me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details visit the Equity Services webpage.

Tentative Class schedule:

IMPORTANT: This schedule is just to give you an overview. Because of several factors, it is quite possible that the timing of topics will be changing as we go further into the course. Some topics may possibly be added and some may be removed. The practice problems lists that will be available on Brightspace will give you the exact topics covered on each class.

Note that we do not have class on May 18 and August 3rd. There is also a pause with classes from June 18 to July 5th (so no classes during that time). Our first class after that break will be on Monday July 6th.

May 6th: Antiderivative, definition of indefinite integral, Basic rules of integration.

May 11th to May 15th: Definite integral, Fundamental Theorem of Calculus, Substitution method

May 18th to May 22nd: Integration by partial fractions, Tabular Method, Integration of trigonometric functions Part 1

May 25th to May 29th: Integration of trigonometric functions Part 2, Integration by trigonometric substitution

June 1st to June 5th: Improper Integrals, Sequences

June 8th to June 12th: Sequences, Series, Convergent or Divergent Series, Telescoping series.

June 15th to June 17th: Geometric series, Integral Test, Comparison Test, Limit Comparison Test

July 6th to July 10th: Absolutely Convergent Series, Ratio Test, Root Test

July 13th to July 17th: Alternating Series Test, Conditionally Convergent Series, Power Series

July 20th to July 24th: Radius of Convergence of a Power Series, Interval of convergence, Representation of a function, Differentiation and Integration of a series.

July 27th to July 31st: Taylor and MacLaurin Series, Binomial Series, Parametric equations of a curve, Tangent line of a parametric curve

August 5: Horizontal and vertical Tangent line, arc length of a curve, polar coordinations, polar curves

August 10th to August 14th: To be determined.